

Proyek ISYS6895003
Sistem Informasi <Retail Fashion Data>

Kelas : BP11

Kelompok : 07

Anggota :

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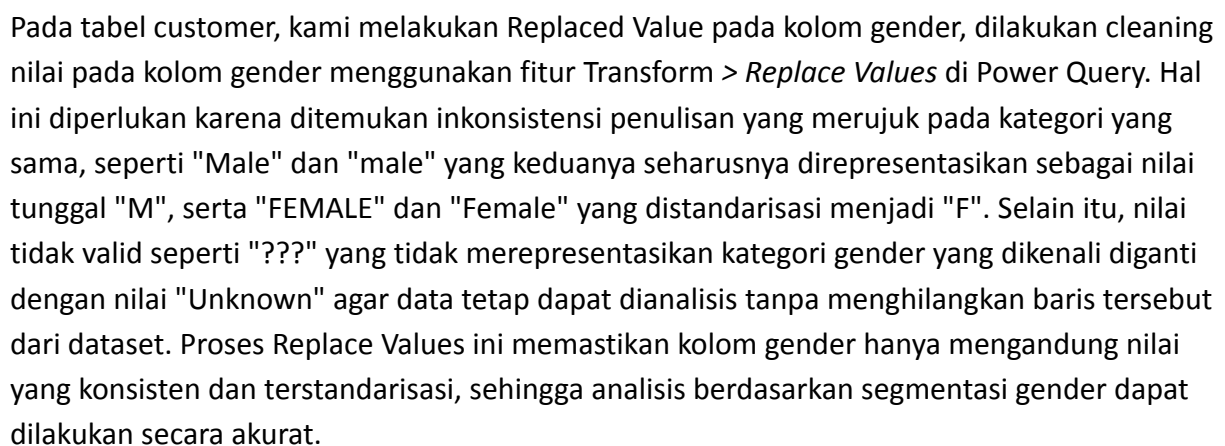
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BINUS UNIVERSITY
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- Manipulating Text



Proses pembersihan data diawali dengan menghapus spasi yang tidak diperlukan pada seluruh kolom bertipe teks menggunakan fitur *Transform > Trim* di Power Query. Langkah ini dilakukan untuk memastikan tidak ada karakter spasi tersembunyi di awal maupun di akhir nilai data yang dapat menyebabkan inkonsistensi saat proses analisis, penggabungan tabel, maupun pemfilteran data. Dengan menerapkan Trim secara menyeluruh pada semua kolom teks, seluruh nilai data menjadi seragam dan bersih dari karakter yang tidak terlihat namun berpotensi mengganggu proses pengolahan data selanjutnya.

Formula Bar: `= Table.ReplaceValue(@"Trimmed Text", "!!!", "Unknown", Replacer.ReplaceText, {"category"})`

product_id	category	color	size	season	supplier	cost_price
P000001	Bottoms	Black	XL	Spring	suppl/erb	7826
P000002	Unknown	Black	M	Summer	suppl/ner	196.7
P000003	Tops	Black	XL	Summer	suppl/ner	2762
P000004	Accessories	White	XL	Winter	suppl/erb	743
P000005	Accessories	Unknown	XL	Fall	suppl/cra	1204
P000006	Shoes	Black	M	Fall	suppl/erc	6869
P000007	Tops	Blue	S	Fall	suppl/erl	1353
P000008	Unknown	Yellow	M	Fall	suppl/erc	5003
P000009	Tops	Yellow	XS	Summer	suppl/erh	1411
P000010	Accessories	Black	S	Winter	suppl/erd	1968
P000011	Bottoms	Green	XS	Summer	suppl/crd	4184
P000012	Tops	Green	XL	Fall	suppl/erh	975
P000013	Accessories	White	S	Fall	suppl/ers	591
P000014	Shoes	Black	XL	Fall	suppl/crd	3507
P000015	Bottoms	Blue	XS	Spring	suppl/erd	17.76
P000016	Shoes	White	M	Spring	suppl/erd	3157
P000017	Bottoms	White	S	Spring	suppl/erb	811
P000018	Accessories	Black	XL	Summer	suppl/era	1949
P000019	Dresses	Black	L	Winter	suppl/erb	2812
P000020	Bottoms	White	XL	Spring	suppl/erb	6805
P000021	Shoes	White	XS	Spring	suppl/erb	5435
P000022	Accessories	White	XS	Spring	suppl/erb	7405
P000023	Bottoms	Red	M	Spring	suppl/erd	5923
P000024	Dresses	White	S	Summer	suppl/cra	1735
P000025	Dresses	Green	L	Spring	suppl/erh	4807
P000026	Tops	Yellow	XS	Fall	suppl/erd	4324
P000027	Tops	Blue	S	Winter	suppl/era	5373
P000028	Shoes	Blue	L	Spring	suppl/erh	1111
P000029	Bottoms	Yellow	M	Fall	suppl/erd	6417
P000030	Bottoms	Black	S	Fall	suppl/erd	2834

Kami juga melakukan Replaced Value pada kolom category yang awalnya terdapat ??? kami ubah semua menjadi Unknown, setelah itu juga tetep melakukan trimmed text agar tidak ada spasi yang berlebih.

- Working with Dates

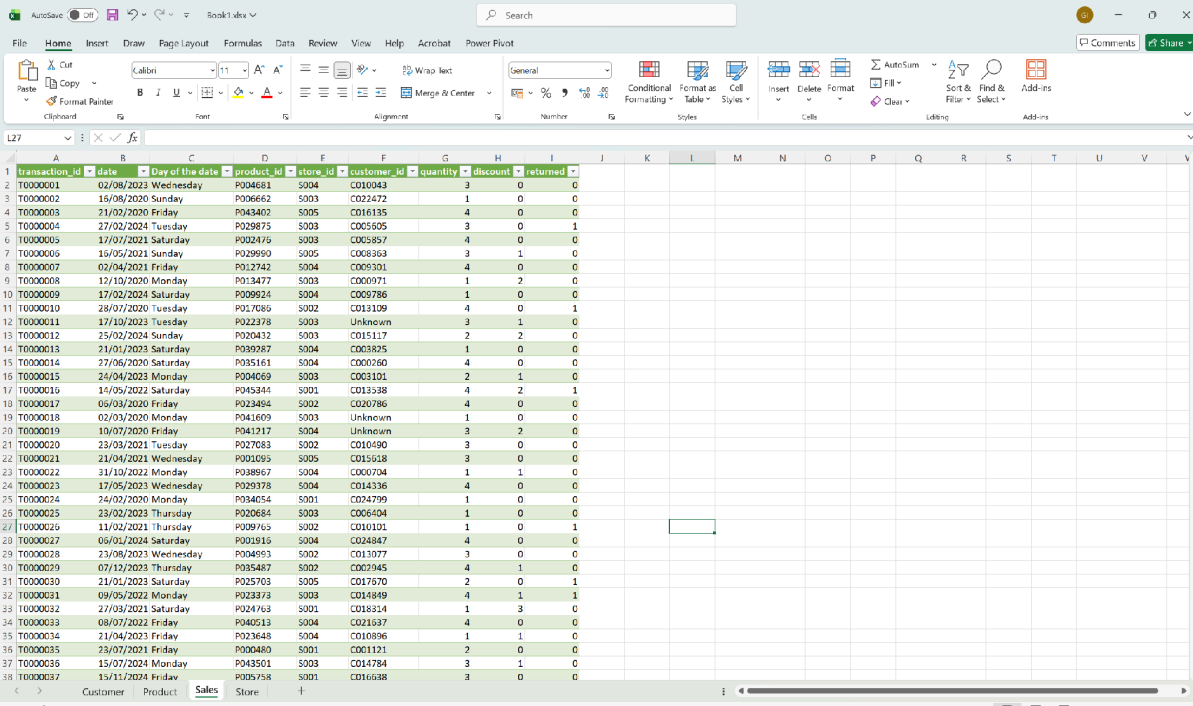
Formula Bar: `= Table.ReorderColumns(*Renamed Columns,{"transaction_id", "date", "day of the date", "product_id", "store_id", "customer_id", "quantity"})`

transaction_id	date	day of the date	product_id	store_id	customer_id	quantity
T0000001	02/08/2022	Wednesday	P004981	S004	C002043	
T0000002	16/08/2020	Sunday	P000662	S003	C022472	
T0000003	21/02/2020	Friday	P043402	S005	C006135	
T0000004	27/02/2024	Tuesday	P022875	S003	C005905	
T0000005	17/07/2021	Saturday	P002476	S003	C005857	
T0000006	16/05/2021	Sunday	P002990	S005	C008363	
T0000007	02/04/2021	Friday	P012742	S004	C003061	
T0000008	12/10/2020	Monday	P013477	S003	C000971	
T0000009	17/02/2024	Saturday	P009924	S004	C009786	
T0000010	28/07/2020	Tuesday	P017086	S002	C013109	
T0000011	12/10/2021	Monday	P011108	S003	Unknown	
T0000012	28/02/2024	Sunday	P000482	S003	C005117	
T0000013	21/01/2023	Saturday	P003287	S004	C000325	
T0000014	11/06/2020	Saturday	P010161	S004	C000140	
T0000015	24/04/2023	Monday	P004609	S003	C001101	
T0000016	14/05/2022	Saturday	P013341	S001	C013338	
T0000017	06/11/2021	Monday	P011494	S007	C001096	
T0000018	02/03/2020	Monday	P016609	S003	Unknown	
T0000019	10/07/2020	Friday	P012117	S004	Unknown	
T0000020	21/01/2021	Monday	P010061	S007	C010490	
T0000021	21/04/2021	Wednesday	P010695	S005	C005518	
T0000022	31/10/2022	Monday	P038967	S004	C000704	
T0000023	17/05/2023	Wednesday	P020378	S004	C014386	
T0000024	24/02/2020	Monday	P034054	S001	C024799	
T0000025	28/02/2023	Thursday	P020694	S003	C006004	
T0000026	11/02/2021	Thursday	P00765	S002	C001061	
T0000027	06/11/2024	Saturday	P010116	S004	C010447	
T0000028	29/08/2023	Wednesday	P004993	S002	C003077	
T0000029	07/12/2023	Thursday	P035487	S002	C002045	
T0000030	21/01/2023	Saturday	P027703	S005	C007570	
T0000031	00/02/2023	Monday	P013372	S003	C010300	

Pada tahap *working with dates*, dilakukan pengolahan data tanggal transaksi untuk menghasilkan informasi tambahan berupa nama hari transaksi. Proses ini dilakukan menggunakan fitur Date → Day → Name of Day pada Power Query. Hasil pengolahan tersebut menghasilkan data seperti Monday, Tuesday, Wednesday, dan seterusnya berdasarkan tanggal transaksi yang tersedia pada dataset.

Penambahan informasi hari transaksi bertujuan untuk mempermudah analisis pola penjualan berdasarkan hari tertentu. Dengan adanya atribut ini, perusahaan dapat mengetahui kapan aktivitas transaksi cenderung lebih tinggi atau lebih rendah. Informasi tersebut dapat digunakan sebagai dasar dalam pengambilan keputusan bisnis, seperti penentuan strategi promosi, pengelolaan stok produk, maupun evaluasi performa penjualan pada hari tertentu. Serta juga tetap melakukan trim agar tidak ada spasi yang berlebih.

2.Named Ranged and Tables



Transaction_Id	date	Day of the date	product_id	store_id	customer_id	quantity	discount	returned
T0000001	02/08/2023	Wednesday	P004681	S004	C010043	3	0	0
T0000002	15/08/2020	Sunday	P006662	S003	C022472	1	0	0
T0000003	21/02/2020	Friday	P043402	S005	C016135	4	0	0
T0000004	27/02/2024	Tuesday	P029875	S003	C005505	3	0	1
T0000005	17/07/2021	Saturday	P002476	S003	C005957	4	0	0
T0000006	16/09/2024	Sunday	P009990	S005	C003363	3	1	0
T0000007	02/04/2023	Friday	P012742	S004	C003301	4	0	0
T0000008	12/10/2020	Monday	P013477	S003	C000971	1	2	0
T0000009	17/02/2024	Saturday	P009924	S004	C009786	1	0	0
T0000010	28/07/2020	Tuesday	P017086	S002	C013109	4	0	1
T0000011	17/10/2023	Tuesday	P022378	S003	Unknown	3	1	0
T0000012	25/02/2024	Sunday	P020432	S003	C015117	2	2	0
T0000013	21/01/2023	Saturday	P039387	S004	C003025	1	0	0
T0000014	27/06/2020	Saturday	P035161	S004	C000260	4	0	0
T0000015	24/04/2023	Monday	P004069	S003	C003101	2	1	0
T0000016	14/05/2022	Saturday	P045844	S001	C013338	4	2	1
T0000017	06/03/2020	Friday	P023494	S002	C020786	4	0	0
T0000018	02/03/2020	Monday	P041609	S003	Unknown	1	0	0
T0000019	19/07/2020	Friday	P041217	S004	Unknown	3	2	0
T0000020	23/03/2021	Tuesday	P027083	S002	C010490	3	0	0
T0000021	21/04/2021	Wednesday	P001095	S005	C015018	3	0	0
T0000022	31/10/2022	Monday	P038967	S004	C000704	1	1	0
T0000023	17/05/2023	Wednesday	P029378	S004	C014336	4	0	0
T0000024	24/02/2020	Monday	P034054	S001	C024799	1	0	0
T0000025	23/02/2023	Thursday	P020684	S003	C006404	1	0	0
T0000026	11/02/2021	Thursday	P009765	S002	C010101	1	0	1
T0000027	06/02/2024	Saturday	P001916	S004	C024847	4	0	0
T0000028	23/08/2023	Wednesday	P004953	S002	C011077	3	0	0
T0000029	07/12/2023	Thursday	P035487	S002	C002945	4	1	0
T0000030	21/01/2023	Saturday	P025703	S005	C017670	2	0	1
T0000031	09/09/2022	Monday	P023373	S003	C014849	4	1	1
T0000032	27/03/2021	Saturday	P024763	S001	C018314	1	3	0
T0000033	08/07/2022	Friday	P040513	S004	C021837	4	0	0
T0000034	21/04/2023	Friday	P023648	S004	C010896	1	1	0
T0000035	23/07/2021	Friday	P000480	S001	C001121	2	0	0
T0000036	15/07/2024	Monday	P043501	S003	C014784	3	1	0
T0000037	15/11/2024	Friday	P005758	S001	C016638	3	0	0

Semua table sudah dalam bentuk excel table.

Excel screenshot showing a data table with columns: transaction_id, date, Day of the date, product_id, store_id, customer_id, quantity, discount, returned. A 'New Name' dialog box is open, showing the name 'quantity' and the scope 'Workbook'. The formula bar shows '=sales_coun_2quantity()'. The status bar at the bottom indicates 'Average: 2.557091342 Count: 49490 Sum: 124076'.

Excel screenshot showing the same data table. The formula bar shows '=SUM(quantity)'. A tooltip for the SUM function is visible, showing the formula '=SUM(number1, [number2], ...)'. The status bar at the bottom indicates 'Average: 2.557091342 Count: 49490 Sum: 124076'.

✕ ✓ <i>fx</i> =SUM(quantity)			
	C	D	E
	Price Change (%)	0%	
	Total Quantity Sold	124076	
	Skenario	124076	
	-15%	105464,6	
	-10%	111668,4	
	-5%	117872,2	
	0%	124076	
	5%	130279,8	
	10%	136483,6	
	15%	142687,4	

1. Quantity

Named Range QuantityData dibuat berdasarkan kolom quantity pada tabel penjualan (sales_messy). Kolom ini berisi jumlah produk yang dibeli dalam setiap transaksi.

Named Range ini digunakan untuk menghitung total produk yang terjual, melakukan analisis jumlah pembelian pelanggan, serta mendukung pembuatan visualisasi dashboard terkait tren penjualan produk.

Contoh penggunaan formula:

=SUM(Quantity)

Formula tersebut digunakan untuk menghitung total keseluruhan produk yang berhasil terjual.

Transaction_id	Date	Day of the date	Product_id	Store_id	Customer_id	Quantity	Discount	Returned
T0000001	02/08/2023	Wednesday	P004681	S004	C010043	3	0	0
T0000002	16/08/2020	Sunday	P006662	S003	C022472	1	0	0
T0000003	21/02/2020	Friday	P043402	S005	C016135	4	0	0
T0000004	27/02/2024	Tuesday	P029875	S003	C005505	3	0	1
T0000005	17/07/2021	Saturday	P002476	S003	C005857	4	0	0
T0000006	16/09/2021	Sunday	P029990	S005	C006368	3	1	0
T0000007	02/04/2021	Friday	P012742	S004	C009301	4	0	0
T0000008	12/10/2020	Monday	P013477	S003	C000971	1	2	0
T0000009	17/02/2024	Saturday	P009924	S004	C009786	1	0	0
T0000010	28/07/2020	Tuesday	P017086	S002	C013109	4	0	1
T0000011	17/10/2023	Tuesday	P022378	S003	Unknown	3	1	0
T0000012	25/02/2024	Sunday	P000432	S003	C003117	2	2	0
T0000013	21/01/2023	Saturday	P030267	S004	C003255	1	0	0
T0000014	27/06/2020	Saturday	P035161	S004	C000260	4	0	0
T0000015	24/04/2023	Monday	P004069	S003	C003101	2	1	0
T0000016	14/05/2022	Saturday	P045344	S001	C013538	4	2	1
T0000017	06/03/2020	Friday	P023494	S002	C020786	4	0	0
T0000018	02/03/2020	Monday	P041609	S003	Unknown	1	0	0
T0000019	10/07/2020	Friday	P041217	S004	Unknown	3	2	0
T0000020	23/03/2021	Tuesday	P027083	S002	C010490	3	0	0
T0000021	21/04/2021	Wednesday	P001055	S005	C015518	3	0	0
T0000022	31/10/2022	Monday	P038967	S004	C000704	1	1	0
T0000023	17/05/2023	Wednesday	P029378	S004	C014336	4	0	0
T0000024	24/02/2020	Monday	P034054	S001	C024799	1	0	0
T0000025	23/02/2023	Thursday	P020684	S003	C006404	1	0	0
T0000026	11/02/2021	Thursday	P009765	S002	C010101	1	0	1
T0000027	06/02/2024	Saturday	P001915	S004	C024947	4	0	0
T0000028	23/08/2023	Wednesday	P004999	S002	C013077	3	0	0
T0000029	07/12/2023	Thursday	P035487	S002	C002945	4	1	0
T0000030	21/01/2023	Saturday	P025703	S005	C017670	2	0	1
T0000031	09/05/2022	Monday	P023373	S003	C014849	4	1	1
T0000032	27/03/2021	Saturday	P024763	S001	C018314	1	3	0
T0000033	08/07/2022	Friday	P040513	S004	C021837	4	0	0
T0000034	21/04/2023	Friday	P023648	S004	C010996	1	1	0
T0000035	23/07/2021	Friday	P000480	S001	C001121	2	0	0
T0000036	15/07/2024	Monday	P043501	S003	C014784	3	1	0
T0000037	15/11/2024	Friday	P005758	S001	C016638	3	0	0

2. Returned

Named Range ReturnedData dibuat berdasarkan kolom returned pada tabel penjualan (sales_messy). Kolom ini menggunakan tipe data biner, yaitu:

- 0 = produk tidak dikembalikan
- 1 = produk dikembalikan

Named Range ini digunakan untuk menganalisis jumlah transaksi pengembalian barang (return) dan membantu perusahaan memahami tingkat retur produk.

Contoh penggunaan formula:

=SUM(Returned)

Formula tersebut digunakan untuk menghitung total transaksi produk yang dikembalikan oleh pelanggan.

3.Logical, Lookup, Conditional Functions

The screenshot displays the Microsoft Excel interface with a data table. The formula bar at the top shows the logical function: `=IF([@returned];"Returned";"Not Returned")`. The table contains 38 rows of transaction data, including transaction ID, date, day of the week, product ID, store ID, customer ID, quantity, discount, returned status, returned info, and category.

transaction_id	date	Day of the date	product_id	store_id	customer_id	quantity	discount	returned	returned_info	category
T0000001	02/08/2023	Wednesday	P004661	S004	C010043	3	0	0	Not Returned	Shoes
T0000002	16/08/2020	Sunday	P006663	S003	C027472	1	0	0	Not Returned	Accessories
T0000003	21/02/2020	Friday	P043402	S005	C016135	4	0	0	Not Returned	Shoes
T0000004	27/02/2024	Tuesday	P029875	S003	C005605	3	0	1	Returned	Shoes
T0000005	17/07/2021	Saturday	P002476	S003	C005857	4	0	0	Not Returned	Tops
T0000006	16/05/2021	Sunday	P029900	S005	C008363	3	1	0	Not Returned	Accessories
T0000007	02/04/2021	Friday	P012742	S004	C009301	4	0	0	Not Returned	Bottoms
T0000008	12/10/2020	Monday	P013477	S003	C009071	1	2	0	Not Returned	Accessories
T0000009	17/02/2024	Saturday	P009924	S004	C009786	1	0	0	Not Returned	Tops
T0000010	28/07/2020	Tuesday	P017066	S002	C013109	4	0	1	Returned	Dresses
T0000011	17/10/2023	Tuesday	P022378	S003	Unknown	3	1	0	Not Returned	Accessories
T0000012	25/02/2024	Sunday	P020432	S003	C015117	2	2	0	Not Returned	Shoes
T0000013	21/01/2023	Saturday	P039287	S004	C003825	1	0	0	Not Returned	Tops
T0000014	27/06/2020	Saturday	P035161	S004	C000260	4	0	0	Not Returned	Tops
T0000015	12/10/2020	Monday	P004068	S003	C003101	1	0	0	Not Returned	Accessories
T0000016	14/05/2022	Saturday	P045344	S001	C013538	4	2	1	Returned	Tops
T0000017	06/03/2020	Friday	P023404	S002	C020786	2	0	0	Not Returned	Tops
T0000018	02/03/2020	Monday	P041609	S003	Unknown	1	0	0	Not Returned	Tops
T0000019	10/07/2020	Friday	P041217	S004	Unknown	3	2	0	Not Returned	Shoes
T0000020	23/03/2021	Tuesday	P027083	S002	C010490	3	0	0	Not Returned	Bottoms
T0000021	21/04/2021	Wednesday	P001095	S005	C015528	3	0	0	Not Returned	Tops
T0000022	31/10/2022	Monday	P039867	S004	C000704	1	1	0	Not Returned	Tops
T0000023	17/06/2023	Wednesday	P029378	S004	C014336	4	0	0	Not Returned	Dresses
T0000024	24/02/2020	Monday	P034054	S001	C024799	1	0	0	Not Returned	Tops
T0000025	23/02/2023	Thursday	P020684	S003	C006404	1	0	0	Not Returned	Bottoms
T0000026	11/02/2021	Thursday	P009765	S002	C010101	1	0	1	Returned	Dresses
T0000027	06/01/2024	Saturday	P001916	S004	C024847	4	0	0	Not Returned	Accessories
T0000028	23/09/2023	Wednesday	P004093	S002	C011077	3	0	0	Not Returned	Accessories
T0000029	07/13/2023	Thursday	P035487	S002	C007045	4	1	0	Not Returned	Shoes
T0000030	21/01/2023	Saturday	P025703	S005	C017670	2	0	1	Returned	Tops
T0000031	09/09/2022	Monday	P023373	S003	C014849	4	1	1	Returned	Tops
T0000032	27/03/2021	Saturday	P024763	S001	C018314	1	3	0	Not Returned	Shoes
T0000033	08/07/2022	Friday	P040513	S004	C021637	4	0	0	Not Returned	Bottoms
T0000034	21/04/2023	Friday	P023648	S004	C010936	1	1	0	Not Returned	Bottoms
T0000035	23/07/2021	Friday	P000480	S001	C001121	2	0	0	Not Returned	Accessories
T0000036	15/07/2024	Monday	P043501	S003	C014784	3	1	0	Not Returned	Shoes
T0000037	15/11/2024	Friday	P005758	S001	C016638	3	0	0	Not Returned	Accessories

Logical Function (IF)

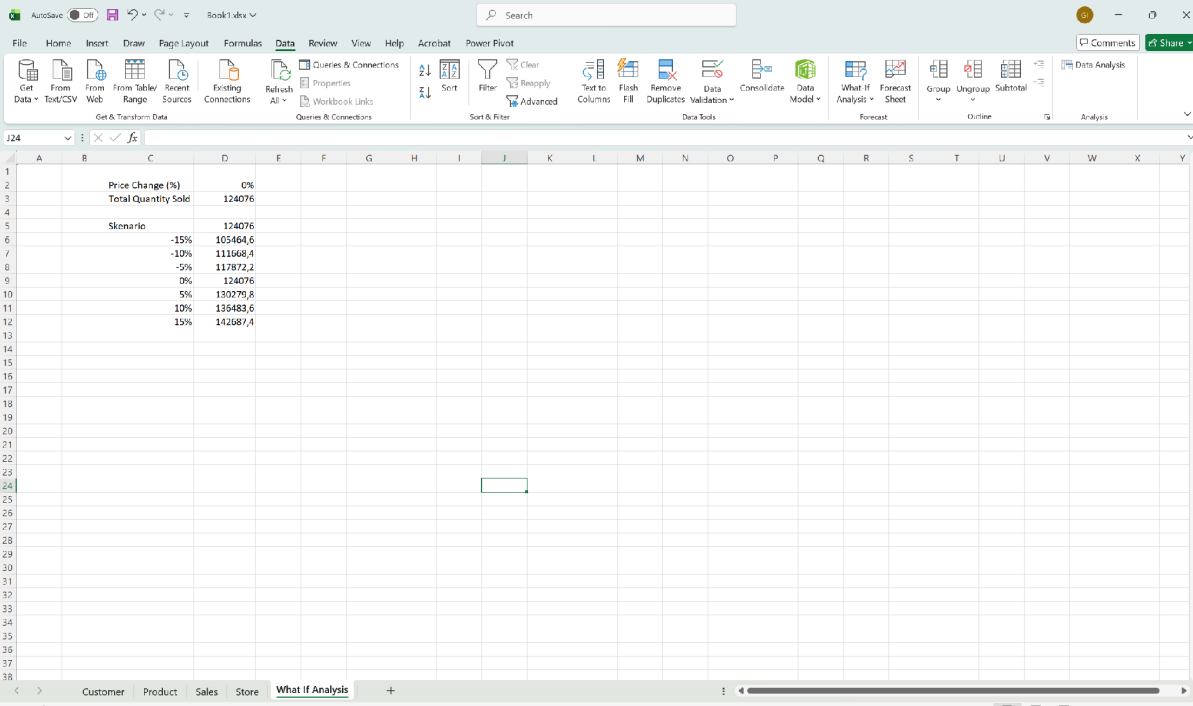
Pada proyek ini digunakan fungsi logika IF untuk mengubah data numerik pada kolom *returned* menjadi informasi yang lebih mudah dipahami. Kami melakukan logical function pada kolom baru bernama *returned_info*. Formula tersebut digunakan untuk mengecek nilai pada kolom *returned*. Jika nilai sama dengan 1, maka Excel akan menampilkan teks “Returned”. Jika nilainya 0, maka akan ditampilkan “Not Returned”. Fungsi ini membantu proses analisis data menjadi lebih mudah dibaca dibandingkan hanya menggunakan angka biner. Hasil dari fungsi ini kemudian digunakan dalam proses analisis transaksi dan visualisasi dashboard terkait pengembalian produk.

transaction_id	date	day_of_the_date	product_id	store_id	customer_id	quantity	discount	returned	returned_info	category
T0000001	02/08/2023	Wednesday	P004681	S004	C010043	3	0	0	Not Returned	Shoes
T0000002	16/08/2020	Sunday	P006662	S003	C022472	1	0	0	Not Returned	Accessories
T0000003	21/02/2020	Friday	P043402	S005	C016135	4	0	0	Not Returned	Shoes
T0000004	27/02/2024	Tuesday	P029875	S003	C005605	3	0	1	Returned	Shoes
T0000005	17/07/2021	Saturday	P002476	S003	C005857	4	0	0	Not Returned	Tops
T0000006	16/05/2021	Sunday	P029960	S005	C006353	3	1	0	Not Returned	Accessories
T0000007	03/04/2021	Friday	P013742	S004	C009301	4	0	0	Not Returned	Bottoms
T0000008	12/10/2020	Monday	P013477	S003	C000971	1	2	0	Not Returned	Accessories
T0000009	17/02/2024	Saturday	P009924	S004	C009786	1	0	0	Not Returned	Tops
T0000010	26/07/2020	Tuesday	P017086	S002	C013109	4	0	1	Returned	Dresses
T0000011	17/10/2023	Tuesday	P022378	S003	Unknown	3	1	0	Not Returned	Accessories
T0000012	25/02/2024	Sunday	P010432	S003	C015117	2	2	0	Not Returned	Shoes
T0000013	24/04/2023	Saturday	P039287	S004	C003825	1	0	0	Not Returned	Tops
T0000014	27/04/2020	Saturday	P035161	S004	C000260	4	0	0	Not Returned	Tops
T0000015	24/04/2023	Monday	P004069	S003	C003101	2	1	0	Not Returned	Accessories
T0000016	14/05/2022	Saturday	P045344	S001	C013538	4	2	1	Returned	Tops
T0000017	06/03/2020	Friday	P023404	S002	C020786	4	0	0	Not Returned	Tops
T0000018	02/03/2020	Monday	P041609	S003	Unknown	1	0	0	Not Returned	Tops
T0000019	10/07/2020	Friday	P041217	S004	Unknown	3	2	0	Not Returned	Shoes
T0000020	23/04/2021	Tuesday	P017083	S002	C004990	3	0	0	Not Returned	Bottoms
T0000021	21/04/2021	Wednesday	P001095	S005	C015618	3	0	0	Not Returned	Tops
T0000022	31/10/2022	Monday	P038967	S004	C000704	1	1	0	Not Returned	Tops
T0000023	17/05/2023	Wednesday	P029378	S004	C014336	4	0	0	Not Returned	Dresses
T0000024	24/02/2020	Monday	P034054	S001	C024799	1	0	0	Not Returned	Tops
T0000025	23/02/2023	Thursday	P020684	S003	C006404	1	0	0	Not Returned	Bottoms
T0000026	11/02/2021	Thursday	P009765	S002	C010101	1	0	1	Returned	Dresses
T0000027	06/04/2024	Saturday	P001916	S004	C024647	4	0	0	Not Returned	Accessories
T0000028	23/08/2023	Wednesday	P004963	S002	C013077	3	0	0	Not Returned	Accessories
T0000029	07/12/2023	Thursday	P035487	S002	C002945	4	1	0	Not Returned	Shoes
T0000030	21/01/2023	Saturday	P025703	S005	C017670	2	0	1	Returned	Tops
T0000031	09/05/2022	Monday	P023373	S003	C014849	4	1	1	Returned	Tops
T0000032	27/03/2021	Saturday	P024763	S001	C018314	1	3	0	Not Returned	Shoes
T0000033	06/07/2022	Friday	P040513	S004	C021837	4	0	0	Not Returned	Bottoms
T0000034	21/04/2023	Friday	P013648	S004	C010596	1	1	0	Not Returned	Bottoms
T0000035	23/07/2021	Friday	P000480	S001	C001121	2	0	0	Not Returned	Accessories
T0000036	15/07/2024	Monday	P043501	S003	C014784	3	1	0	Not Returned	Shoes
T0000037	15/11/2024	Friday	P005758	S001	C016638	3	0	0	Not Returned	Accessories

Lookup Function (VLOOKUP)

Pada proyek ini digunakan fungsi VLOOKUP untuk mengambil informasi kategori produk dari tabel produk ke tabel penjualan berdasarkan product_id. Formula tersebut digunakan untuk mencari nilai *product_id* pada tabel produk, kemudian mengambil data kategori produk yang sesuai seperti Accessories, Shoes, Tops, Bottoms, dan Dresses. Fungsi VLOOKUP membantu proses penggabungan data antar tabel sehingga informasi kategori produk dapat digunakan dalam analisis penjualan dan pembuatan dashboard.

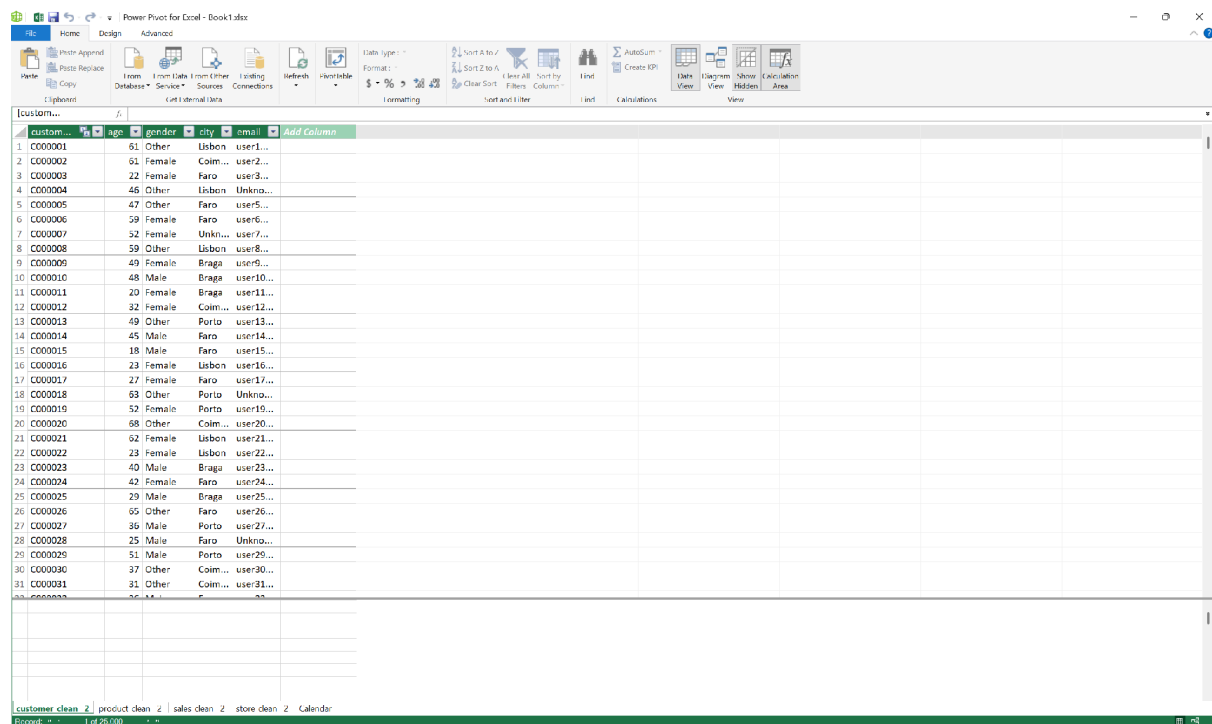
4. Analysis with Data Tables



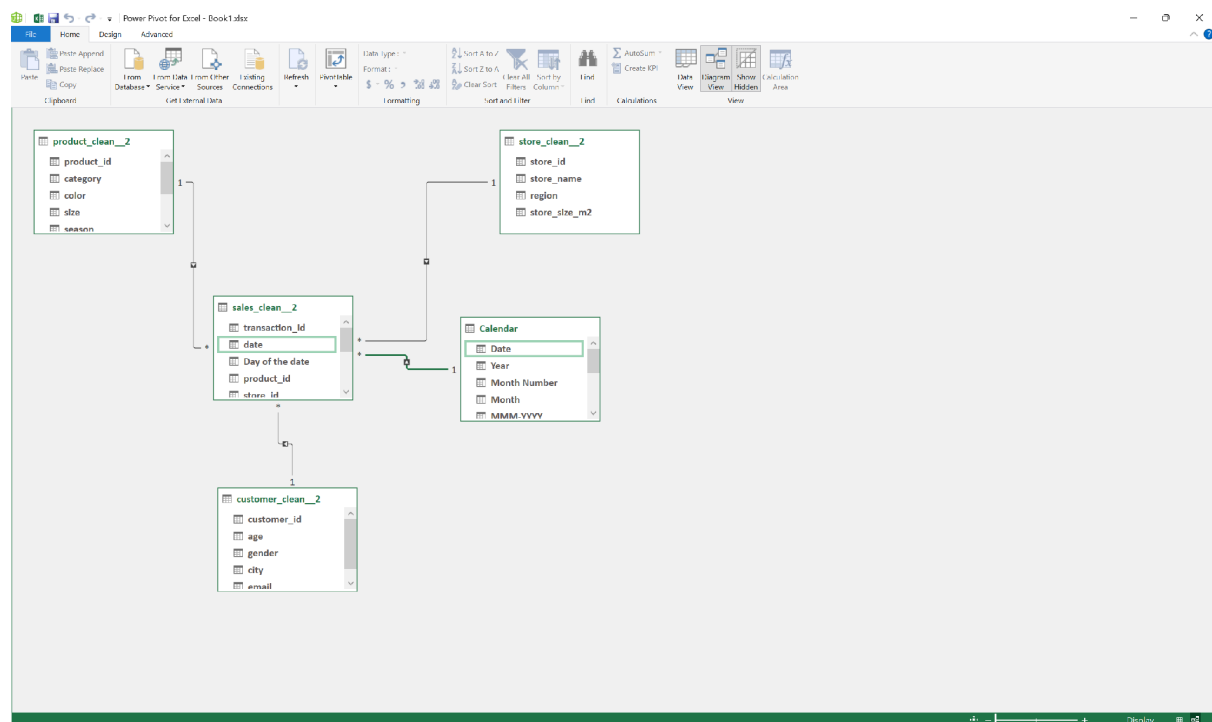
Price Change (%)	Total Quantity Sold
0%	124076
-15%	105464,6
-10%	111568,4
-5%	117872,2
0%	124076
5%	130279,8
10%	136483,6
15%	142687,4

What-If Analysis dilakukan untuk menganalisis dampak perubahan persentase penjualan terhadap total quantity sold pada dataset. Analisis ini menggunakan fitur Data Table pada Microsoft Excel dengan beberapa skenario perubahan, yaitu -15%, -10%, -5%, 0%, 5%, 10%, dan 15%. Nilai awal total quantity sold berdasarkan dataset adalah sebesar 124.076. Selanjutnya dilakukan simulasi menggunakan formula perhitungan projected quantity sold untuk melihat perubahan hasil penjualan pada setiap skenario. Hasil analisis menunjukkan bahwa ketika terjadi penurunan sebesar 15%, total quantity sold ikut menurun menjadi sekitar 105.464. Sebaliknya, ketika terjadi peningkatan sebesar 15%, projected quantity sold meningkat menjadi sekitar 142.687. Dari hasil tersebut dapat disimpulkan bahwa perubahan persentase penjualan memiliki pengaruh langsung terhadap jumlah produk yang terjual. Semakin besar peningkatan persentase penjualan, maka quantity sold juga meningkat, sedangkan penurunan persentase menyebabkan jumlah penjualan menurun. Analisis ini berguna untuk membantu proses perencanaan dan pengambilan keputusan bisnis berdasarkan kemungkinan perubahan kondisi penjualan di masa mendatang.

5. Power Pivot and The Data Model



	custom	age	gender	city	email	Add Column
1	C000001	61	Other	Lisbon	user1...	
2	C000002	61	Female	Coim...	user2...	
3	C000003	22	Female	Faro	user3...	
4	C000004	46	Other	Lisbon	Unkno...	
5	C000005	47	Other	Faro	user5...	
6	C000006	59	Female	Faro	user6...	
7	C000007	32	Female	Unkno...	user7...	
8	C000008	59	Other	Lisbon	user8...	
9	C000009	49	Female	Braga	user9...	
10	C000010	48	Male	Braga	user10...	
11	C000011	29	Female	Braga	user11...	
12	C000012	32	Female	Coim...	user12...	
13	C000013	49	Other	Porto	user13...	
14	C000014	45	Male	Faro	user14...	
15	C000015	18	Male	Faro	user15...	
16	C000016	23	Female	Lisbon	user16...	
17	C000017	27	Female	Faro	user17...	
18	C000018	63	Other	Porto	Unkno...	
19	C000019	52	Female	Porto	user19...	
20	C000020	68	Other	Coim...	user20...	
21	C000021	62	Female	Lisbon	user21...	
22	C000022	23	Female	Lisbon	user22...	
23	C000023	40	Male	Braga	user23...	
24	C000024	42	Female	Faro	user24...	
25	C000025	29	Male	Braga	user25...	
26	C000026	66	Other	Faro	user26...	
27	C000027	36	Male	Porto	user27...	
28	C000028	25	Male	Faro	Unkno...	
29	C000029	51	Male	Porto	user29...	
30	C000030	37	Other	Coim...	user30...	
31	C000031	31	Other	Coim...	user31...	



Pada tahap ini dilakukan proses pengolahan data menggunakan fitur Power Pivot pada Microsoft Excel. Langkah pertama yang dilakukan adalah mengaktifkan Power Pivot agar dataset dapat dimasukkan ke dalam Data Model. Setelah data dimasukkan ke dalam Data Model, dilakukan pembuatan relationship antar tabel menggunakan kolom yang memiliki keterkaitan.

Power Pivot for Excel - Book1.xlsx

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=CUBEVALUE("ThisWorkbookDataModel";"[Measures].[Total Returned]")

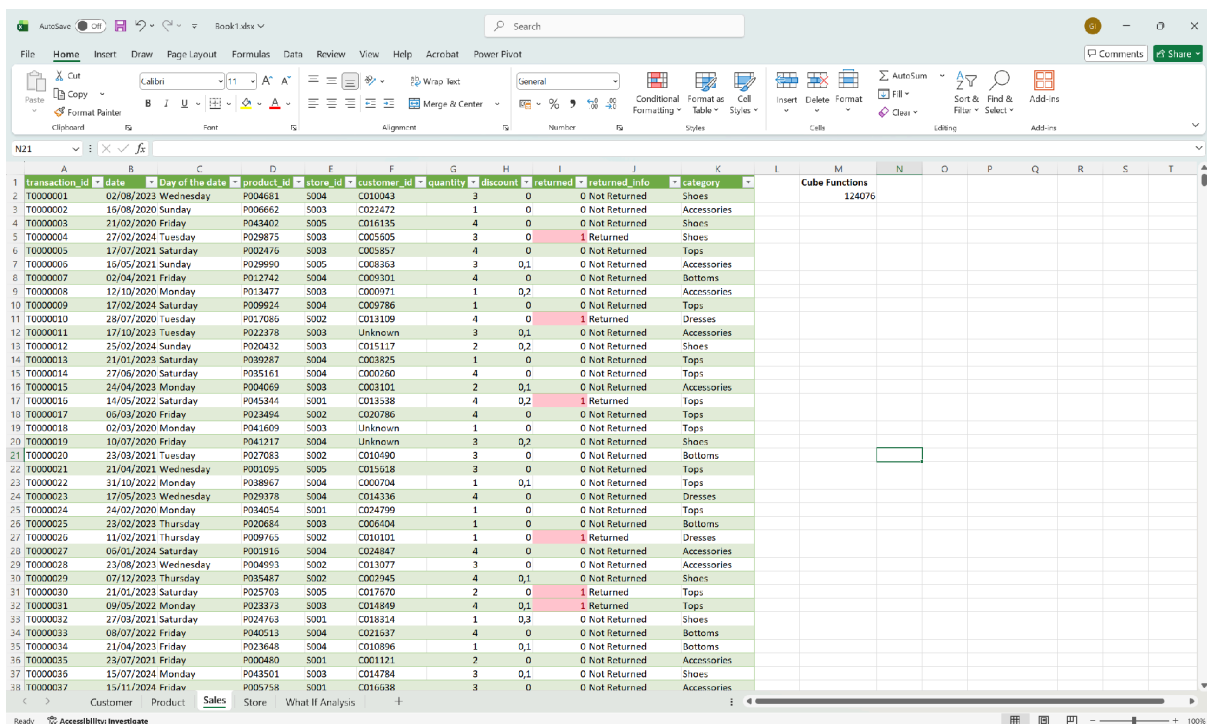
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
1				Average Age																				
2				42,46128																				
3																								

Selain membuat relationship, dilakukan juga pembuatan calculated field atau measure menggunakan bahasa DAX (Data Analysis Expressions). Measure pertama adalah Total Quantity Sold yang digunakan untuk menghitung total jumlah produk yang terjual menggunakan fungsi SUM pada kolom quantity. Measure kedua adalah Total Returned yang digunakan untuk menghitung jumlah produk yang mengalami return berdasarkan kolom returned yang berisi nilai 0 dan 1. Measure ini membantu dalam menganalisis performa penjualan dan tingkat pengembalian produk.

Selanjutnya digunakan Cube Function untuk mengambil hasil perhitungan dari Power Pivot Data Model ke worksheet Excel biasa. Fungsi yang digunakan adalah CUBEVALUE, yang berfungsi menampilkan hasil measure secara langsung ke worksheet sehingga mempermudah proses analisis dan visualisasi data.

6. Conditional and Number Formatting

- Conditional Formatting



transaction_id	date	day_of_the_week	product_id	store_id	customer_id	quantity	discount	returned	returned_info	category
T0000001	02/08/2023	Wednesday	P004651	S004	C010043	3	0	0	0 Not Returned	Shoes
T0000002	16/08/2020	Sunday	P006662	S003	C023472	1	0	0	0 Not Returned	Accessories
T0000003	21/03/2020	Friday	P043402	S005	C016135	4	0	0	0 Not Returned	Shoes
T0000004	27/02/2024	Tuesday	P029875	S003	C005505	3	0	1	1 Returned	Shoes
T0000005	17/01/2021	Saturday	P002476	S003	C005857	4	0	0	0 Not Returned	Tops
T0000006	16/05/2021	Sunday	P029990	S005	C008363	3	0,1	0	0 Not Returned	Accessories
T0000007	02/04/2021	Friday	P012742	S004	C000301	4	0	0	0 Not Returned	Bottoms
T0000008	12/10/2020	Monday	P013477	S003	C000971	1	0,2	0	0 Not Returned	Accessories
T0000009	17/02/2024	Saturday	P009924	S004	C009786	1	0	0	0 Not Returned	Tops
T0000010	28/01/2020	Tuesday	P017086	S002	C013109	4	0	1	1 Returned	Dresses
T0000011	17/10/2023	Tuesday	P022378	S003	Unknown	3	0,1	0	0 Not Returned	Accessories
T0000012	25/02/2024	Sunday	P020432	S003	C015117	2	0,2	0	0 Not Returned	Shoes
T0000013	21/01/2023	Saturday	P030987	S004	C003825	1	0	0	0 Not Returned	Tops
T0000014	27/04/2020	Saturday	P035161	S004	C000760	4	0	0	0 Not Returned	Tops
T0000015	24/04/2023	Monday	P004069	S003	C003101	2	0,1	0	0 Not Returned	Accessories
T0000016	14/05/2022	Saturday	P045344	S001	C013538	4	0,2	1	1 Returned	Tops
T0000017	06/03/2022	Friday	P023494	S002	C020786	4	0	0	0 Not Returned	Tops
T0000018	02/03/2020	Monday	P041609	S003	Unknown	1	0	0	0 Not Returned	Tops
T0000019	10/07/2020	Friday	P041217	S004	Unknown	3	0,2	0	0 Not Returned	Shoes
T0000020	23/03/2021	Tuesday	P027083	S002	C010490	3	0	0	0 Not Returned	Bottoms
T0000021	23/04/2021	Wednesday	P031095	S005	C013538	3	0	0	0 Not Returned	Tops
T0000022	31/10/2022	Monday	P038967	S004	C000704	1	0,1	0	0 Not Returned	Tops
T0000023	17/05/2023	Wednesday	P029378	S004	C014336	4	0	0	0 Not Returned	Dresses
T0000024	24/02/2020	Monday	P034054	S001	C024799	1	0	0	0 Not Returned	Tops
T0000025	23/02/2023	Thursday	P020684	S003	C006404	1	0	0	0 Not Returned	Bottoms
T0000026	11/02/2021	Thursday	P009765	S002	C010101	1	0	1	1 Returned	Dresses
T0000027	06/03/2024	Saturday	P001919	S004	C024847	4	0	0	0 Not Returned	Accessories
T0000028	23/08/2023	Wednesday	P004953	S002	C013077	3	0	0	0 Not Returned	Accessories
T0000029	07/12/2023	Thursday	P035487	S002	C002945	4	0,1	0	0 Not Returned	Shoes
T0000030	21/01/2023	Saturday	P025703	S005	C017670	2	0	1	1 Returned	Tops
T0000031	09/05/2022	Monday	P023373	S003	C014849	4	0,1	1	1 Returned	Tops
T0000032	27/03/2021	Saturday	P024763	S001	C018314	1	0,3	0	0 Not Returned	Shoes
T0000033	08/07/2022	Friday	P040513	S004	C021937	4	0	0	0 Not Returned	Bottoms
T0000034	21/04/2023	Friday	P023648	S004	C010896	1	0,1	0	0 Not Returned	Bottoms
T0000035	23/07/2021	Friday	P000480	S001	C001121	2	0	0	0 Not Returned	Accessories
T0000036	15/07/2024	Monday	P043501	S003	C014784	3	0,1	0	0 Not Returned	Shoes
T0000037	15/11/2024	Friday	P005758	S001	C016638	3	0	0	0 Not Returned	Accessories

Conditional Formatting diterapkan pada kolom returned untuk menyoroti transaksi yang memiliki nilai 1, yang menandakan produk dikembalikan (returned). Aturan formatting dibuat agar sel dengan nilai 1 otomatis diberi warna kuning sehingga transaksi return dapat lebih mudah diidentifikasi saat proses analisis data dilakukan. Penerapan formatting ini membantu meningkatkan keterbacaan data dan mempermudah proses monitoring transaksi pengembalian barang.

- Number Formatting

transaction_id	date	Day of the date	product_id	store_id	customer_id	quantity	discount	returned	returned_info	category
1	02/08/2023	Wednesday	P004681	S004	C010043	3	0	0	0 Not Returned	Shoes
2	16/08/2020	Sunday	P006662	S003	C022472	1	0	0	0 Not Returned	Accessories
3	21/02/2020	Friday	P043402	S005	C016135	4	0	0	0 Not Returned	Shoes
4	27/02/2024	Tuesday	P029875	S003	C005605	3	0	1	1 Returned	Shoes
5	17/07/2021	Saturday	P002476	S003	C005857	4	0	0	0 Not Returned	Tops
6	16/05/2021	Sunday	P029990	S005	C008363	3	0.1	0	0 Not Returned	Accessories
7	02/04/2021	Friday	P012742	S004	C009301	4	0	0	0 Not Returned	Bottoms
8	12/10/2020	Monday	P013477	S003	C000971	1	0.2	0	0 Not Returned	Accessories
9	17/02/2024	Saturday	P009924	S004	C009786	1	0	0	0 Not Returned	Tops
10	28/07/2020	Tuesday	P017086	S002	C013109	4	0	1	1 Returned	Dresses
11	17/10/2023	Tuesday	P022378	S003	Unknown	3	0.1	0	0 Not Returned	Accessories
12	25/02/2024	Sunday	P020432	S003	C015117	2	0.2	0	0 Not Returned	Shoes
13	21/01/2023	Saturday	P039287	S004	C003825	1	0	0	0 Not Returned	Tops
14	27/06/2020	Saturday	P035161	S004	C000260	4	0	0	0 Not Returned	Tops
15	24/04/2023	Monday	P004069	S003	C003101	2	0.1	0	0 Not Returned	Accessories
16	14/05/2022	Saturday	P045344	S001	C013538	4	0.2	1	1 Returned	Tops
17	06/03/2020	Friday	P023484	S002	C020786	4	0	0	0 Not Returned	Tops
18	02/03/2020	Monday	P041609	S003	Unknown	1	0	0	0 Not Returned	Tops
19	10/07/2020	Friday	P041217	S004	Unknown	3	0.2	0	0 Not Returned	Shoes
20	23/03/2021	Tuesday	P027083	S002	C010490	3	0	0	0 Not Returned	Bottoms
21	20/00/2021	Wednesday	P001095	S005	C015618	3	0	0	0 Not Returned	Tops
22	31/10/2022	Monday	P038967	S004	C000704	1	0.1	0	0 Not Returned	Tops
23	17/05/2023	Wednesday	P029378	S004	C014336	4	0	0	0 Not Returned	Dresses
24	24/02/2020	Monday	P034054	S001	C024799	1	0	0	0 Not Returned	Tops
25	23/02/2023	Thursday	P020684	S003	C006404	1	0	0	0 Not Returned	Bottoms
26	11/02/2022	Thursday	P009765	S002	C010101	1	0	1	1 Returned	Dresses
27	06/01/2024	Saturday	P001916	S004	C024847	4	0	0	0 Not Returned	Accessories
28	23/08/2023	Wednesday	P004993	S002	C013077	3	0	0	0 Not Returned	Accessories
29	07/12/2023	Thursday	P035487	S002	C002945	4	0.1	0	0 Not Returned	Shoes
30	21/01/2023	Saturday	P025703	S005	C017670	2	0	1	1 Returned	Tops
31	09/05/2022	Monday	P023373	S003	C014849	4	0.1	1	1 Returned	Tops
32	27/03/2021	Saturday	P024763	S001	C018314	1	0.3	0	0 Not Returned	Shoes
33	08/07/2022	Friday	P040513	S004	C021637	4	0	0	0 Not Returned	Bottoms
34	21/04/2023	Friday	P023648	S004	C010896	1	0.1	0	0 Not Returned	Bottoms
35	23/07/2021	Friday	P000480	S001	C001121	2	0	0	0 Not Returned	Accessories
36	15/07/2024	Monday	P043501	S003	C014784	3	0.1	0	0 Not Returned	Shoes
37	15/11/2024	Friday	P005758	S001	C016638	3	0	0	0 Not Returned	Accessories

transaction_id	date	Day of the date	product_id	store_id	customer_id	quantity	discount	returned	returned_info	category
1	02/08/2023	Wednesday	P004681	S004	C010043	3	0%	0	0 Not Returned	Shoes
2	16/08/2020	Sunday	P006662	S003	C022472	1	0%	0	0 Not Returned	Accessories
3	21/02/2020	Friday	P043402	S005	C016135	4	0%	0	0 Not Returned	Shoes
4	27/02/2024	Tuesday	P029875	S003	C005605	3	0%	1	1 Returned	Shoes
5	17/07/2021	Saturday	P002476	S003	C005857	4	0%	0	0 Not Returned	Tops
6	16/05/2021	Sunday	P029990	S005	C008363	3	10%	0	0 Not Returned	Accessories
7	02/04/2021	Friday	P012742	S004	C009301	4	0%	0	0 Not Returned	Bottoms
8	12/10/2020	Monday	P013477	S003	C000971	1	20%	0	0 Not Returned	Accessories
9	17/02/2024	Saturday	P009924	S004	C009786	1	0%	0	0 Not Returned	Tops
10	28/07/2020	Tuesday	P017086	S002	C013109	4	0%	1	1 Returned	Dresses
11	17/10/2023	Tuesday	P022378	S003	Unknown	3	10%	0	0 Not Returned	Accessories
12	25/02/2024	Sunday	P020432	S003	C015117	2	20%	0	0 Not Returned	Shoes
13	21/01/2023	Saturday	P039287	S004	C003825	1	0%	0	0 Not Returned	Tops
14	27/06/2020	Saturday	P035161	S004	C000260	4	0%	0	0 Not Returned	Tops
15	24/04/2023	Monday	P004069	S003	C003101	2	10%	0	0 Not Returned	Accessories
16	14/05/2022	Saturday	P045344	S001	C013538	4	20%	1	1 Returned	Tops
17	06/03/2020	Friday	P023484	S002	C020786	4	0%	0	0 Not Returned	Tops
18	02/03/2020	Monday	P041609	S003	Unknown	1	0%	0	0 Not Returned	Tops
19	10/07/2020	Friday	P041217	S004	Unknown	3	20%	0	0 Not Returned	Shoes
20	23/03/2021	Tuesday	P027083	S002	C010490	3	0%	0	0 Not Returned	Bottoms
21	20/00/2021	Wednesday	P001095	S005	C015618	3	0%	0	0 Not Returned	Tops
22	31/10/2022	Monday	P038967	S004	C000704	1	10%	0	0 Not Returned	Tops
23	17/05/2023	Wednesday	P029378	S004	C014336	4	0%	0	0 Not Returned	Dresses
24	24/02/2020	Monday	P034054	S001	C024799	1	0%	0	0 Not Returned	Tops
25	23/02/2023	Thursday	P020684	S003	C006404	1	0%	0	0 Not Returned	Bottoms
26	11/02/2022	Thursday	P009765	S002	C010101	1	0%	1	1 Returned	Dresses
27	06/01/2024	Saturday	P001916	S004	C024847	4	0%	0	0 Not Returned	Accessories
28	23/08/2023	Wednesday	P004993	S002	C013077	3	0%	0	0 Not Returned	Accessories
29	07/12/2023	Thursday	P035487	S002	C002945	4	10%	0	0 Not Returned	Shoes
30	21/01/2023	Saturday	P025703	S005	C017670	2	0%	1	1 Returned	Tops
31	09/05/2022	Monday	P023373	S003	C014849	4	10%	1	1 Returned	Tops
32	27/03/2021	Saturday	P024763	S001	C018314	1	30%	0	0 Not Returned	Shoes
33	08/07/2022	Friday	P040513	S004	C021637	4	0%	0	0 Not Returned	Bottoms
34	21/04/2023	Friday	P023648	S004	C010896	1	10%	0	0 Not Returned	Bottoms
35	23/07/2021	Friday	P000480	S001	C001121	2	0%	0	0 Not Returned	Accessories
36	15/07/2024	Monday	P043501	S003	C014784	3	10%	0	0 Not Returned	Shoes
37	15/11/2024	Friday	P005758	S001	C016638	3	0%	0	0 Not Returned	Accessories

Number Formatting diterapkan pada kolom discount dengan mengubah tampilan angka menjadi format persentase (%). Formatting ini dilakukan agar nilai diskon lebih mudah dibaca dan dipahami dalam bentuk persentase, seperti 10%, 20%, dan 30%, dibandingkan angka desimal biasa. Dengan penggunaan format persentase, proses analisis data penjualan menjadi lebih jelas, informatif, dan mudah diinterpretasikan.

- Calculated Fields and Measures

The screenshot shows a Microsoft Excel window titled 'Power Pivot for Excel - Book1.xlsx'. The ribbon is set to 'PivotTable Tools' with the 'Formulas' tab selected. A PivotTable is displayed with the following data:

customer_id	age	gender	city	email
22	23	Female	Lisbon	user22...
23	40	Male	Braga	user23...
24	42	Female	Faro	user24...
25	29	Male	Braga	user25...
26	65	Other	Faro	user26...
27	36	Male	Porto	user27...
28	25	Male	Faro	Unkno...
29	51	Male	Porto	user29...
30	37	Other	Coim...	user30...
31	31	Other	Coim...	user31...
32	36	Male	Faro	user32...
33	39	Male	Porto	user33...
34	26	Male	Lisbon	user34...
35	27	Male	Porto	user35...
36	54	Male	Porto	user36...
37	50	Male	Lisbon	user37...
38	40	Other	Lisbon	Unkno...
39	67	Male	Lisbon	user39...
40	54	Male	Lisbon	user40...
41	21	Other	Faro	user41...
42	52	Other	Coim...	user42...
43	55	Female	Lisbon	user43...
44	35	Male	Faro	user44...
45	51	Other	Coim...	user45...
46	64	Male	Porto	user46...
47	22	Male	Coim...	user47...
48	19	Male	Porto	user48...
49	38	Female	Braga	user49...
50	29	Other	Porto	user50...
51	44	Other	Braga	user51...
52	39	Female	Porto	user52...

Below the data, a PivotTable is shown with the following structure:

	Average Age=AVERAGE(Customer[age])
Average Age	42,46128

Measure Average Age dibuat pada tabel Customer untuk menghitung rata-rata usia pelanggan yang terdapat pada dataset. Perhitungan dilakukan menggunakan fungsi AVERAGE() pada kolom age sehingga sistem dapat menghitung nilai rata-rata umur seluruh customer secara otomatis. Formula yang digunakan adalah AVERAGE(Customer[age]). Measure ini bekerja dengan menjumlahkan seluruh nilai usia pelanggan kemudian membaginya dengan jumlah customer yang tersedia pada data. Hasil dari measure bersifat dinamis, sehingga nilainya dapat berubah sesuai filter atau analisis yang diterapkan pada PivotTable maupun Data Model. Penggunaan measure Average Age membantu dalam memahami karakteristik pelanggan berdasarkan rata-rata usia customer yang melakukan transaksi pada dataset penjualan.

Insert Calculated Field

?

×

Name:

profit

▼

Modify

Formula:

= list_price - cost_price

Delete

Fields:

product_id

category

color

size

season

supplier

cost_price

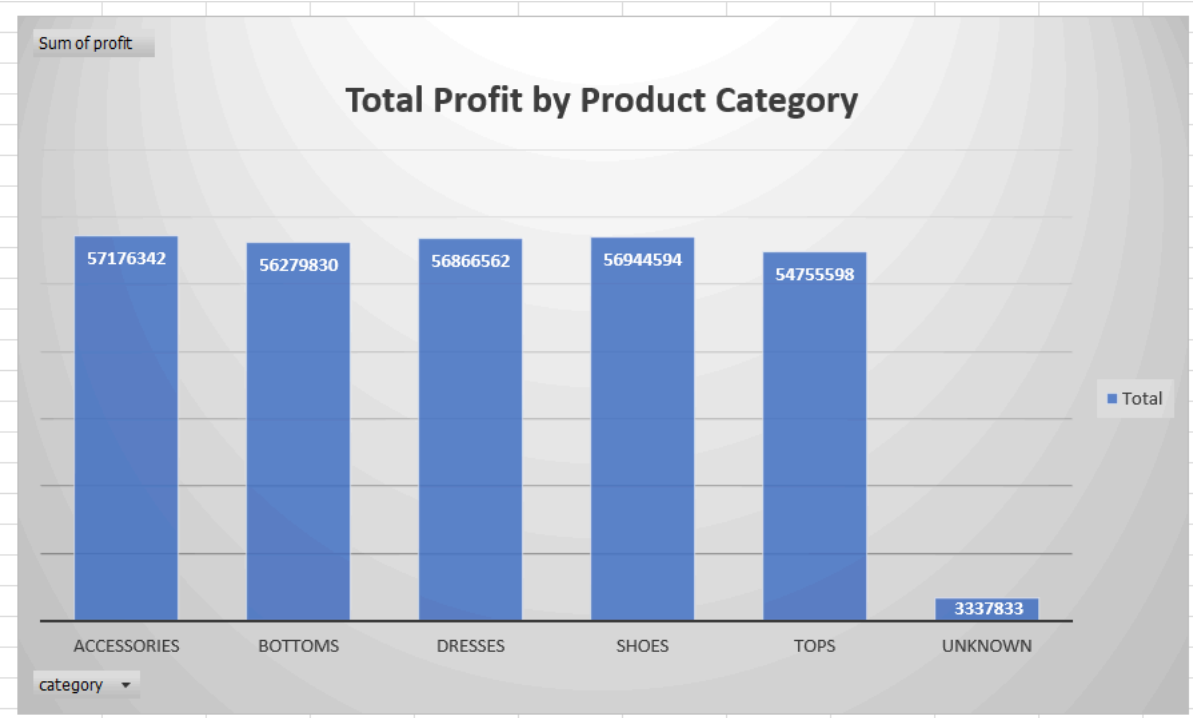
list_price

Insert Field

OK

Close

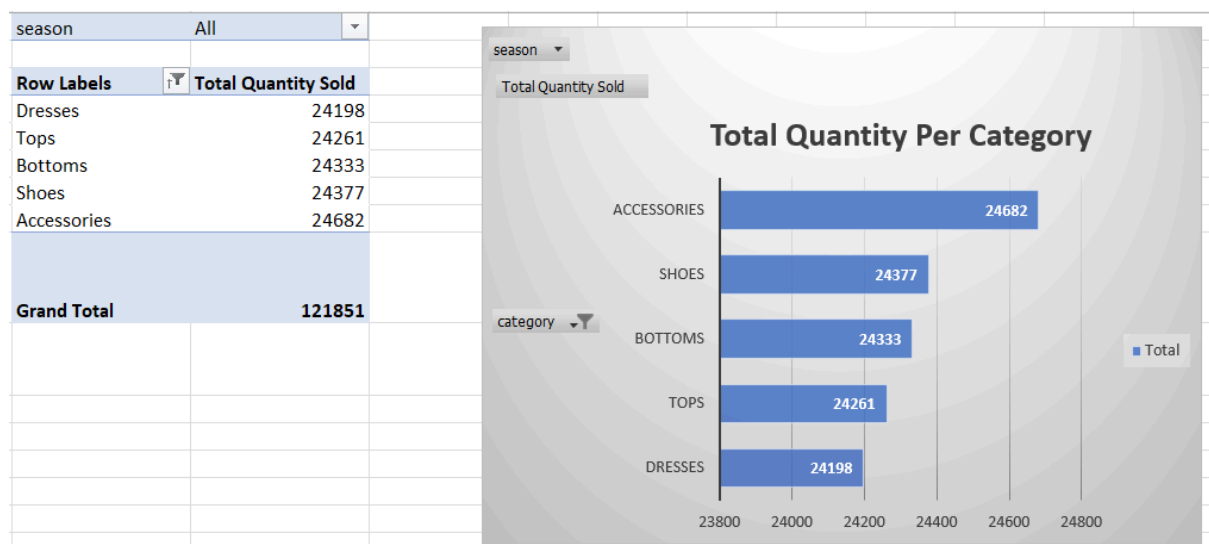
Row Labels	Sum of profit
Accessories	57176342
Bottoms	56279830
Dresses	56866562
Shoes	56944594
Tops	54755598
Unknown	3337833
Grand Total	285360759



Pada analisis ini dibuat sebuah **Calculated Field** bernama **Profit** pada Pivot Table untuk menghitung keuntungan yang diperoleh dari setiap produk. Calculated Field tersebut menggunakan rumus **Profit = List Price – Cost Price**, di mana *List Price* merupakan harga jual produk dan *Cost Price* merupakan biaya perolehan produk. Dengan perhitungan ini, setiap produk memiliki nilai keuntungan yang kemudian dapat dijumlahkan dan dianalisis berdasarkan kategori produk. Penggunaan Calculated Field memungkinkan perhitungan dilakukan secara otomatis di dalam Pivot Table tanpa perlu menambahkan kolom baru pada dataset asli, sehingga proses analisis menjadi lebih efisien dan dinamis.

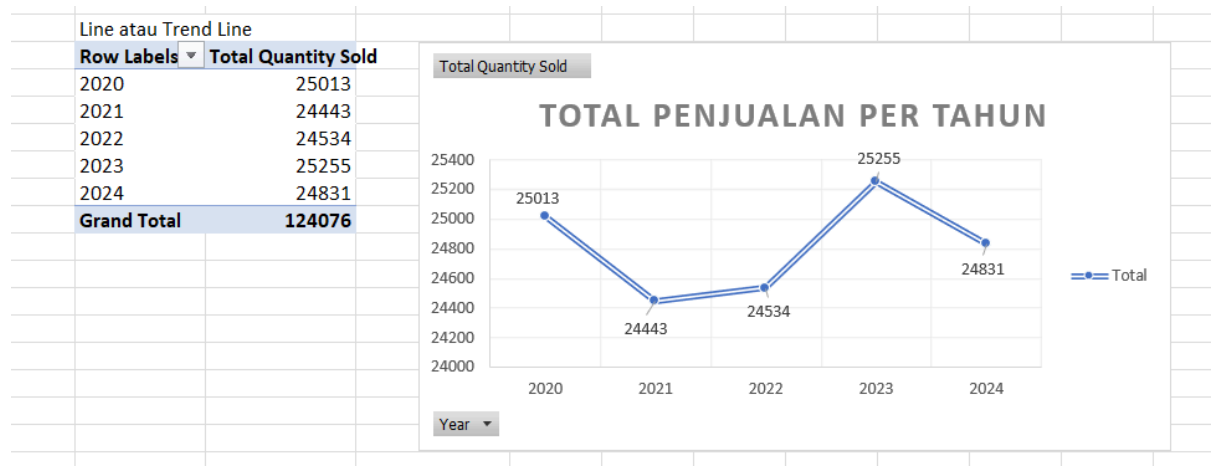
7.Mastering Charts

- Bar Chart



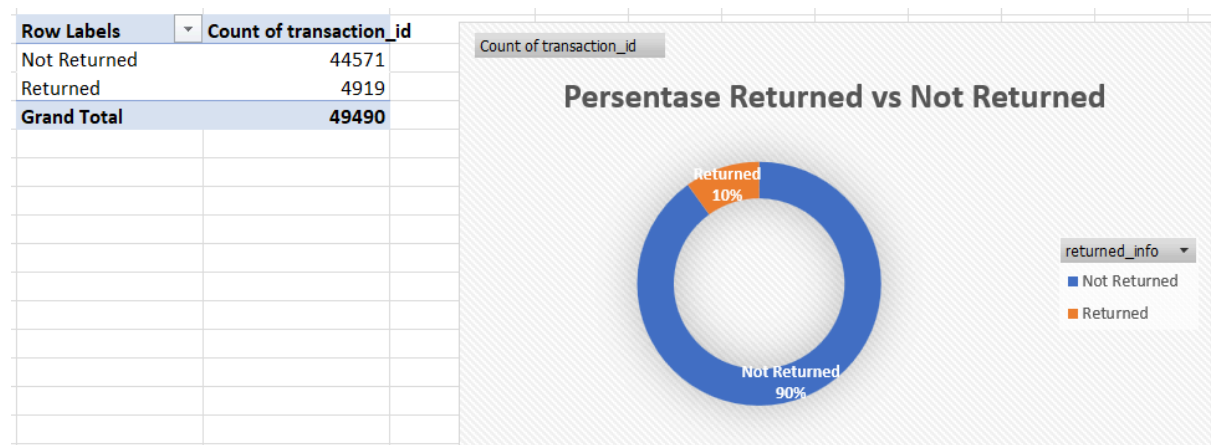
Bar Chart digunakan untuk membandingkan total quantity penjualan pada setiap kategori produk yang terdapat pada dataset. Grafik batang mempermudah visualisasi kategori dengan jumlah penjualan tertinggi dan terendah sehingga perbandingan antar kategori dapat dilihat dengan lebih jelas. Dengan menggunakan chart ini, analisis penjualan produk menjadi lebih mudah dipahami secara visual.

- Line atau Trend Line



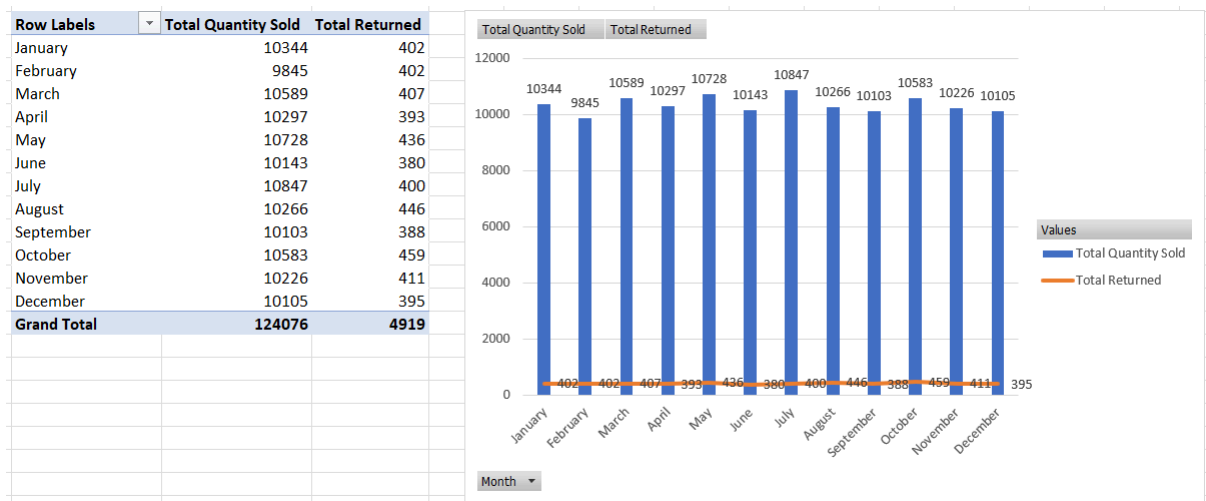
Line Chart digunakan untuk melihat tren penjualan berdasarkan periode waktu tahunan. Grafik garis membantu menunjukkan pola kenaikan maupun penurunan jumlah penjualan dari waktu ke waktu sehingga perkembangan penjualan dapat dianalisis dengan lebih jelas. Visualisasi ini mempermudah identifikasi bulan dengan penjualan tertinggi maupun terendah pada dataset.

- Doughnut Chart



Doughnut Chart digunakan untuk menunjukkan persentase transaksi returned dan not returned pada dataset penjualan. Grafik ini membantu memvisualisasikan proporsi data dalam bentuk persentase sehingga perbandingan antara produk yang direturn dan tidak direturn dapat terlihat dengan lebih mudah. Dari hasil visualisasi terlihat bahwa sebagian besar transaksi termasuk kategori not returned dibandingkan returned.

- Combination Chart



Combination Chart digunakan untuk menggabungkan dua jenis visualisasi dalam satu grafik, yaitu column chart untuk total quantity penjualan dan line chart untuk jumlah returned berdasarkan periode bulanan. Grafik batang menunjukkan total produk yang terjual setiap bulan, sedangkan grafik garis menunjukkan jumlah produk yang direturn pada bulan yang sama. Visualisasi ini membantu menganalisis hubungan antara peningkatan penjualan dengan tingkat returned sehingga pola transaksi dapat dipahami dengan lebih baik.

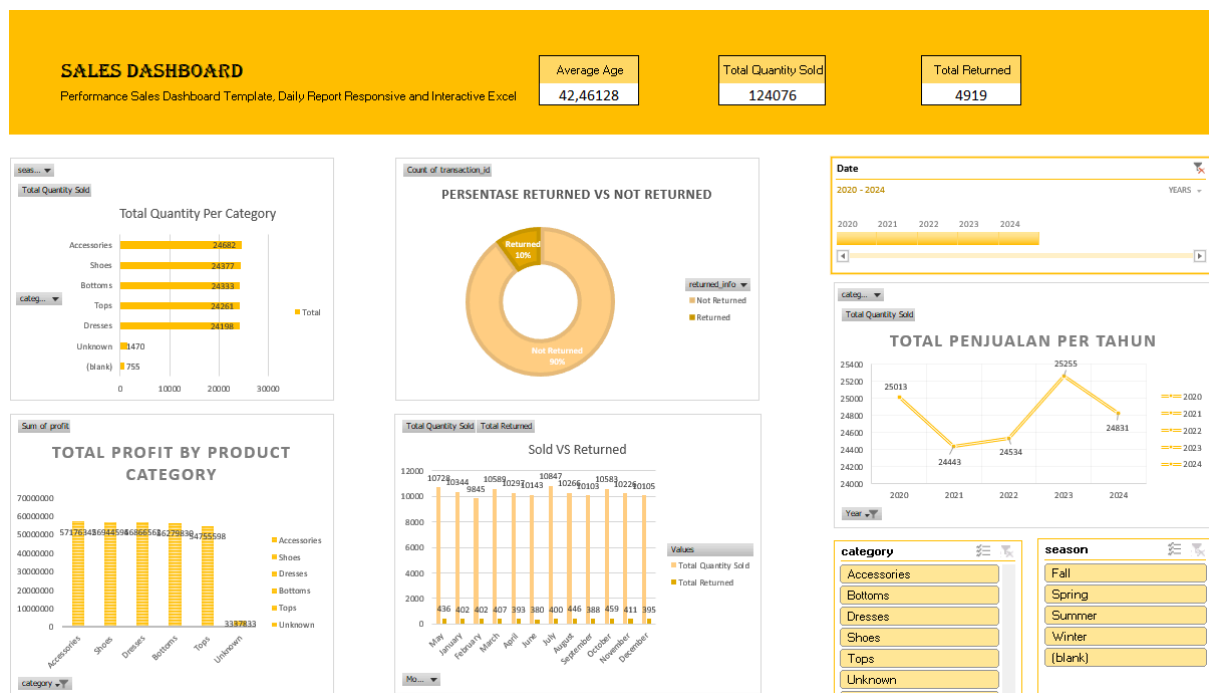
- Column Chart



Grafik **Total Profit by Product Category** menampilkan total keuntungan yang diperoleh dari setiap kategori produk berdasarkan calculated field **Profit**, yang dihitung menggunakan rumus **Profit = List Price – Cost Price**. Rumus ini digunakan untuk mengetahui selisih antara harga jual dan biaya perolehan produk sehingga dapat menggambarkan keuntungan yang

dihasilkan oleh setiap produk. Berdasarkan hasil visualisasi, kategori **Accessories** menghasilkan total profit tertinggi sebesar sekitar 57,18 juta, diikuti oleh kategori Dresses, Shoes, Tops, dan Bottoms yang memiliki nilai profit relatif serupa. Sementara itu, kategori **Unknown** menghasilkan profit yang jauh lebih rendah dibandingkan kategori lainnya. Analisis ini membantu perusahaan mengidentifikasi kategori produk yang memberikan kontribusi keuntungan terbesar sehingga dapat menjadi fokus dalam strategi penjualan dan pengelolaan produk.

- Creating Interactive Dashboard



Dashboard ini dibuat untuk memvisualisasikan dan menganalisis data penjualan perusahaan secara interaktif menggunakan Excel, Power Pivot, Pivot Table, Pivot Chart, Measure, Calculated Field, dan Cube Functions. Tujuan utama dashboard ini adalah membantu pengguna memahami performa penjualan, profit, karakteristik pelanggan, serta tingkat pengembalian produk melalui satu halaman yang mudah dibaca dan dianalisis.

Pada bagian atas dashboard terdapat tiga Key Performance Indicators (KPI) yang ditampilkan menggunakan Cube Functions. KPI pertama adalah **Average Age** sebesar 42,46 tahun yang menunjukkan rata-rata usia pelanggan. KPI kedua adalah **Total Quantity Sold** sebesar 124.076 unit yang menunjukkan total produk yang berhasil terjual. KPI ketiga adalah **Total**

Returned sebesar 4.919 transaksi yang menunjukkan jumlah produk yang dikembalikan oleh pelanggan. Ketiga KPI ini memberikan gambaran umum mengenai kondisi bisnis secara cepat sebelum pengguna melihat detail visualisasi lainnya.

Visualisasi pertama adalah **Total Quantity per Category** yang menunjukkan jumlah produk terjual pada setiap kategori produk. Hasil analisis menunjukkan bahwa kategori Accessories memiliki jumlah penjualan tertinggi, sedangkan kategori Dresses memiliki jumlah penjualan terendah. Grafik ini membantu perusahaan mengetahui kategori produk yang paling diminati pelanggan.

Visualisasi kedua adalah **Total Penjualan per Tahun** yang menampilkan tren penjualan dari tahun 2020 hingga 2024. Dari grafik terlihat bahwa penjualan mengalami penurunan pada tahun 2021, kemudian meningkat hingga mencapai puncaknya pada tahun 2023 sebelum sedikit menurun pada tahun 2024. Grafik ini membantu melihat perkembangan performa bisnis dari tahun ke tahun.

Visualisasi ketiga adalah **Persentase Returned vs Not Returned** yang membandingkan jumlah transaksi yang mengalami pengembalian produk dengan transaksi yang tidak mengalami pengembalian. Hasil analisis menunjukkan bahwa sekitar 90% transaksi tidak mengalami return, sedangkan hanya sekitar 10% transaksi yang direturn. Hal ini menunjukkan bahwa sebagian besar transaksi berjalan dengan baik dan tingkat pengembalian produk relatif rendah.

Visualisasi keempat adalah **Sales Trend dan Returned Trend per Month** yang menggunakan combination chart untuk membandingkan jumlah penjualan dan jumlah return pada setiap bulan. Grafik ini membantu mengidentifikasi apakah peningkatan penjualan pada periode tertentu diikuti oleh peningkatan jumlah produk yang dikembalikan oleh pelanggan.

Visualisasi terakhir adalah **Total Profit by Product Category** yang dibuat menggunakan Calculated Field bernama Profit dengan rumus **List Price – Cost Price**. Grafik ini menunjukkan total keuntungan yang dihasilkan oleh setiap kategori produk. Hasilnya menunjukkan bahwa kategori Accessories memberikan kontribusi profit terbesar, sedangkan kategori Unknown menghasilkan profit paling rendah. Informasi ini dapat digunakan untuk menentukan kategori produk yang paling menguntungkan bagi perusahaan.